NAUTILUS: Auditing Evidence and Exploration in Image Geolocation

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Abstract

NAUTILUS examines localization accuracy, visual evidence, and exploration on 25 European scenes. We compare four static baselines, 11 interactive configurations, and five 15-image masked batches. GLM-5.3-Flash achieves 88.8% of available game points with coordinate tools. However, its repeated runs reveal unstable text interpretation and geographic recall. Masked predictions likewise reveal both robust landmark recognition and dependence on interface labels. These results connect localization quality to visual interpretation, evidence acquisition, and interface control.

1. Introduction

We evaluate multimodal large language models (MLLMs) available in September 2026, building on prior geolocation benchmarks [6, 8, 12]. NAUTILUS connects their predictions to stated visual cues, observed panorama changes, and submitted map pins. Through repeated runs, masked inputs, and an evidence explorer, we therefore examine how cue interpretation and interface control shape localization.

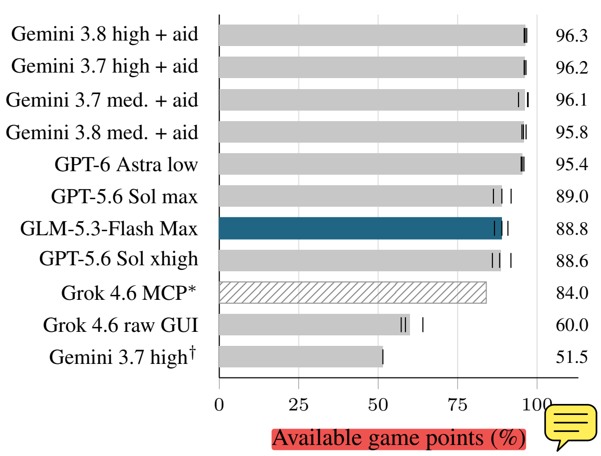

Figure 1. Interactive game points. Bars show means and ticks show individual evaluations (three each, †: one). Grok MCP* combines tier-specific games. Gemini labels denote Flash, with a pin-placement helper for "aid".

2. Method

Scenes and baselines. We curate 25 European OpenGuessr scenes: eight Easy, nine Medium, and eight Hard. PNGs and Street View links preserve starting views and targets. Four pretrained reference systems provide static predictions (Table 1). SALAD retrieves from 2.17 million European OSV-5M references [1] with 1,024 IVF-Flat lists, 64 probes, and top-k filtering. PLONK uses one sample per image and seed 42. Chipoint v2 uses public three-tower fusion and cluster consensus.

Table 1. Static baselines on 25 images. Errors in km, with A_{200} denoting the fraction within 200 km.

Baseline	Mean ↓	Median ↓	A_{200} ↑
PLONK	**501.8**	**167.8**	**56%**
Chipoint v2	536.2	312.1	32%
GeoCLIP	1,183.3	224.2	44%
SALAD + OSV-5M	1,308.5	929.2	20%

Interactive evaluation. We retain 28 complete 25-location evaluations plus 75 Grok MCP tier rounds. Harnesses and reasoning levels follow Figure 1. Gemini's aided variants use a pin-placement helper. GLM uses ZCode and ten Model Context Protocol (MCP) tools for screenshots and coordinate actuation. It has 300 s per round, reports five cues before exploration, and should submit with 60 s remaining. Play excludes external search, metadata, hidden targets, and DOM/network inspection.

Masked-image comparison. Five configurations receive 14 starting images with road-label overlays covered and one with provider metadata covered, in one batch each. We compare their guesses and cue lists with the recorded initial beliefs for the original starting views, before exploration and map refinement. Table 2 compares countries across 182 initial records. A match requires one preferred country. Where initial coordinates are recorded, we also compare geographic error on the corresponding scenes. Provider metadata is analysed separately.

Evidence and metrics. We calculate WGS84 geodesic errors, retaining displayed distances for Astra's online runs. GLM traces and 268 screenshots support action-level analysis, while other-model reports add decision notes. We summarize errors, distance thresholds, and game-point fractions.

Figure 2. Saved GLM screenshots. N444 (left) shows forward movement before a country revision to Belgium. The Greek scene (right) remains unchanged after a drag despite a reported improvement in reading. Road labels are Google-rendered overlays.

3. Results

Localization accuracy and control. Gemini 3.8 high/aided reaches 96.3% of available points, while Astra low reaches 95.4%. PLONK leads the static mean and median measures. Meanwhile, GLM achieves 88.8%, with 138.0 km mean and 22.5 km median error across 75 predictions. Its point fraction falls from 99.7% on Easy to 77.4% on Hard.

Controller behavior also matters. Grok's raw-GUI report recognizes Bastille yet records a pin 979.9 km away. Its MCP Easy runs reach 99.7% and report agreement between intended and submitted pins across all 24 rounds. In contrast, Sol Max's Pula-to-Rome mistake faithfully transfers a wrong landmark belief. Thus both cue interpretation and pin transfer explain errors. Moreover, Astra's Lithuania report records an 85 m pin while describing its farm match as unverified. Its stated uncertainty therefore remains relevant even after a precise result.

Table 2. Country matches on the same 14 road-label scenes. Initial beliefs use three repeats, or two for Sol. Masked guesses use one batch. U counts unresolved initial country choices.

Configuration	Initial	U	Masked
Astra low	39/42	3	13/14
GLM Max	31/42	2	13/14
Sol xhigh	22/28	2	11/14
Sol Max	25/28	0	12/14
Grok xhigh	37/42	1	11/14

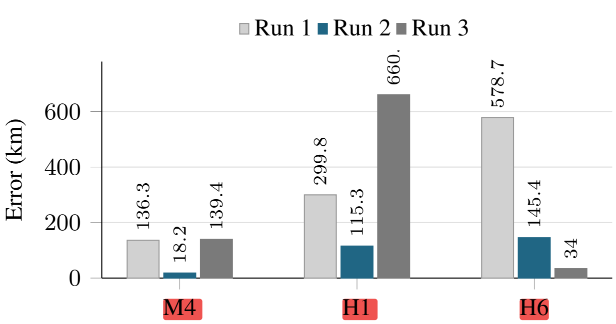

Figure 3. GLM's run-to-run final-pin error variation: N444 exploration (M4), Greek-script reinterpretation (H1), and conflicting road-number associations (H6). Values are rounded displayed distances.

Which cues remain useful? Table 2 compares guesses directly. All five configurations still locate masked Bastille within 110 m, citing the July Column and transit sign. On the Greek-road scene, however, 11 of 13 initial beliefs identify Greece, while all five masked guesses choose other countries, with errors of 454–2,331 km. Their notes instead emphasize Mediterranean vegetation and limestone, showing how the script supplies a more specific geographic cue.

The effect also depends on how a cue is interpreted. GLM initially places N444 in the Netherlands in all three runs, whereas its masked guess favors Belgium from housing and overhead wiring. Likewise, brick architecture leads its masked Bologna guess away from the Milan hypotheses in two initial runs. Across the 14 scenes, its mean initial error, averaged within each scene, is 134.1 km versus 94.0 km masked. Separately, its three initial Finland beliefs become a Norway guess after provider metadata is covered, with 895.2 km error.

Why repeated runs diverge. Figure 3 separates three sources of variation. First, GLM's N444 Run 1 gains new views through forward movement (Figure 2). Houses and overhead infrastructure shift its belief from the Netherlands to Belgium, yielding 18.2 km error against 136.3 and 139.4 km in the other runs. Second, Run 3 reads the Greek overlay as Bulgarian/Cyrillic and claims improved legibility after an ineffective drag. The unchanged view therefore supports an interpretation error, with a final pin 660.2 km away. Third, the same 3426 overlay evokes Finland, Latvia, and Lithuania across runs, producing 578.7, 145.4, and 34.0 km errors. Thus variation arises through evidence acquisition, text interpretation, and geographic recall.

Conclusion. MLLMs localize distinctive scenes well, while ambiguity exposes unstable cue associations. Computer use constrains Grok and unaided Gemini, whereas Astra demonstrates more effective interface handling. Moreover, GLM's 88.8% point fraction shows the potential of open-weight models with suitable tools. Exploration helps when successful actions reveal useful evidence. Limited compute constrained this study to 25 European scenes. Future work should extend geographic coverage, evaluate more models and reasoning levels, and repeat the starting-view comparisons.

A. Related Work

MLLM geolocation benchmarks. LLMGeo finds that fine-tuning helps open models approach closed-model country accuracy [12]. IMAGEO-Bench evaluates coordinate errors, geographic bias, and structured explanations across three image collections [6]. Its results associate successful localization with identifiable landmarks and urban context. These broad evaluations provide the starting point for our closer study of contemporary game-playing agents.

Reasoning and cue reliability. EarthWhere scores the use of annotated visual clues alongside country- and street-level answers [8]. GeoRC compares explanations against 800 expert chains for 500 scenes and documents geographic misattribution, hallucinated cues, and claims of unavailable tool use [9]. We extend this evidence-based assessment to recorded camera transitions and submitted pins, tracing how observations become geographic hypotheses and actions.

Exploration and tool use. WanderBench supplies navigable panorama graphs, while GeoAoT acquires additional views through planned actions [14]. ERGeoBench compares single-view, panorama, and sequential yaw/pitch/zoom conditions, exposing difficulties in metric localization and cross-view consistency [13]. Meanwhile, LocationAgent verifies evidence through a Reasoner–Executor–Recorder hierarchy [7], and Thinking with Map trains agents to query map APIs [5]. With external lookup restricted, NAUTILUS examines camera-control failures and the transfer of a model's intended location to its submitted map pin.

Cue masking and faithfulness. In a controlled comparison, IMAGEO-Bench masks signage, storefront names, and license-plate text, finding substantial streetscape accuracy losses [6]. Similarly, Durrieu et al. test attribution-guided object regions using deletion and insertion [3]. Our masks target Google-rendered road labels and imagery-provider metadata, while traces record which remaining cues models cite. Because language-model explanations can omit influences on their answers [10], we interpret those statements together with the observed views, actions, and coordinates.

Specialized reference systems. GeoCLIP aligns images with GPS representations [11], SALAD aggregates features for retrieval [4], and PLONK models geographic uncertainty generatively [2]. Alongside OSV-5M reference imagery [1], these methods provide the static context for our agent-focused analysis.

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